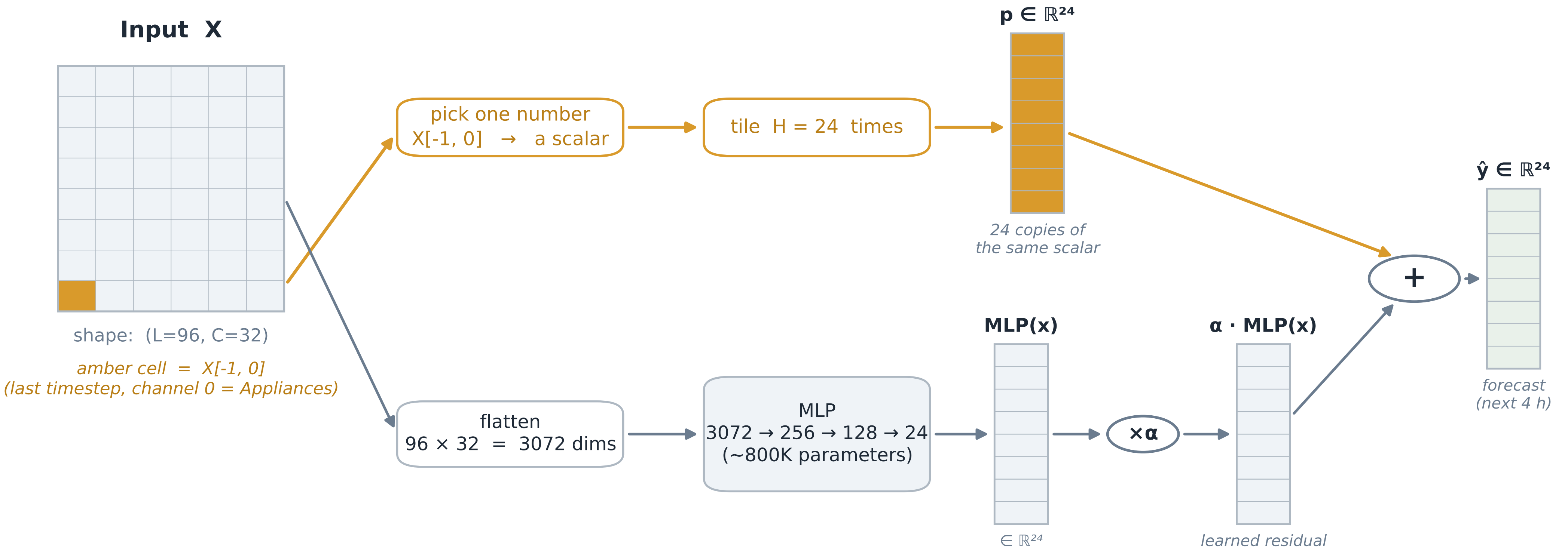
Residual-anchored MLP: $\hat{y}$ = persistence(x) + α · MLP(x)

persistence gives a sensible floor; the MLP only has to learn the small residual; α controls how much the MLP is trusted



α = 0 ⇒ $\hat{y}$ = p (pure persistence; MLP disabled). α = 1 ⇒ $\hat{y}$ = p + MLP(x) (full residual). Adam learns the best α for the data.

Why this beats vanilla MLP: a plain MLP with weight decay drifts toward predicting the mean (≈ 0 after standardisation), which is a bad guess for a spiky signal. With the anchor, the MLP only has to learn the small residual — a far easier problem.